

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	02/07/2026
Team ID	
Project Name	Book Store
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed Book Store solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Form (Email & Password confirmation) • Registration through Gmail Integration • Registration through LinkedIn Integration
FR-2	User Confirmation	• Confirmation via Verification Email link • Confirmation via Mobile One-Time Password (OTP)
FR-3	Book Search & Catalog	• Keyword Search by Title, Author, or ISBN • Advanced Filter by Genre, Price Range, and Condition (New/Used) • Real-time Book Stock Availability View
FR-4	Shopping Cart & Checkout	• Add, update, or remove books in Cart with active session hold • Secure Multi-Gateway Payment Processing (Card, Net Banking, UPI) • Automated digital Invoice generation and email receipt delivery

Non-functional Requirements

Following are the non-functional requirements of the proposed Book Store solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface must be completely responsive across web views and mobile touch devices, featuring an accessible text-to-scale layout design ensuring page-load ease.
NFR-2	Security	All customer authentication operations and transmission pipelines must enforce end-to-end HTTPS/TLS data encryption and securely store passwords using standard hashing algorithms.
NFR-3	Reliability	The data storefront database transactions must operate with full ACID-compliance to ensure zero loss of customer billing data or inventory states during transaction errors.
NFR-4	Performance	Catalog text search query listings and homepage loading routines must execute and populate within 2.0 seconds under nominal network operating thresholds.
NFR-5	Availability	The platform core services should maintain a minimum of 99.9% application uptime schedule yearly, excluding scheduled low-traffic database index maintenance intervals.
NFR-6	Scalability	The application system routing must scale seamlessly to handle a concurrent traffic spike of up to 10,000 active customer browsing sessions during peak bookstore sales.